A Novel Method for Intelligent EMC Management Using a "Knowledge Base"

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***Abstract*—Management of electromagnetic compatibility (EMC) is a key aspect of product manufacturing. The fierce industry competition and numerous regulations (guidelines) make EMC architects, engineers, and manufacturers increasingly anxious to apply EMC management in each phase (design, validation, development, production, deployment, and monitoring) by depending mainly on their experience. To organize the knowledge, regulations, and implementations of EMC management, this study introduces a new knowledge base with a collection of entities, their properties, and a rich set of relationships among them. Then, a machine learning-based framework for EMC management automation and intelligence is proposed. Convolutional neural network is applied to establish and classify the entities, and the knowledge-based question-answering approach is used for question answering. Finally, the EMC management plan is generated automatically. Two experiments were designed to evaluate the proposed method, and results show that both good recall and precision are achieved in the comprehensive EMC management scenarios. This method is useful for the practical EMC management of industrial products and can enhance product quality, shorten the production cycle, and reduce production cost.**

***Index Terms*—Electromagnetic compatibility (EMC), knowledge base (KB), machine learning, management.**

I. Introduction

Management of electromagnetic compatibility (EMC) attracted attention because of concerns about the man-made side-effects of new electromagnetic sources in the early 1960s. The investigations about how to impose the EMC regulations in the production design, acquisition, development, and operations started in the United States Army at the same time [1]. Aside from human protection, the target also involved efforts to eliminate the effect of electromagnetic interference on complex products and projects. In the 1970s and 1980s, the U.S. military published a series of regulations for radiated emission limits and electromagnetic inference (EMI) testing to prevent a critical interference path to integrated systems or products [2].

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Subsequently, the EMC management was applied in various areas. Methodologies flourished for different phases and domains, especially for the industrial products and information equipment [3]–[8]. At the same time, the following challenges proliferated [9]:

1) With the increase in applications and system complexity, EMC management officers sometimes experienced burnout because of their monitoring tasks and the monotonous nature of their work.
2) The education cost of EMC and EMI for engineers became increasingly higher as electronic devices were widely used in all possible areas.
3) The social division of labor renders EMC monitoring in each phase impossible.

Numerous management rules and systems were proposed to solve the problems [10]–[12], but most of them were extremely difficult to implement for two reasons:

1) Most management programs are only processes. Any process without automation requires a complex education and monitoring infrastructure for support, making the cost unacceptable for small business groups, and delaying the projects.
2) Rule-based systems are limited to parameter matching, which cannot apprehend device design and make particular management suggestions for EMI testing and EMC design.

Fortunately, the deep learning technique has made considerable progress in recent years, and it significantly enhanced the industrial intelligence [13], [14]. In this study, we first apply such a technique to EMC management, thereby promoting intelligent and automatic management. The proposed method effectively solves the aforementioned challenges of EMC management in industry.

The entity-based approach is considered as an effective method to structuralize a knowledge domain and is widely used in recommender systems to provide suggestions based on authoritative knowledge [15], [16]. Extensive efforts have been conducted to extract, link, and store entities into a knowledge base (KB) with the rapid development of computer software and hardware techniques [17], [18].

This study introduces a new knowledge structure-based EMC management framework. The management work is translated to a question-answering (QA) system. Moreover, the hierarchical semantic entity extractions by convolutional neural network (CNN) are applied to the EMC designs and procurements. After

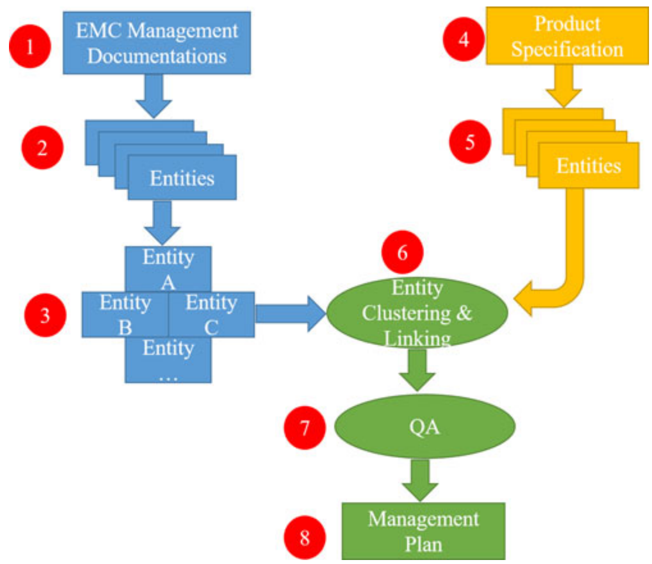

Fig. 1. Phases of overall architecture.

entity linking and ranking, EMC management plans are developed for given EMC scenarios.

Analysis of the published literature indicates that this work is the first attempt to use the knowledge graph to help identify, understand, implement, and monitor the management of EMC for a product or project. Building a KB improves the convenience of EMC management and makes it an automatic process. An evaluation method is also proposed to evaluate the effectiveness of the framework.

The rest of this paper is structured as follows. Section II introduces the main methods applied in our KB. Experimental data and results are shown in Section III. The conclusion is discussed in Section IV.

II. Method Description

In this section, we aim to introduce the overall framework and methods applied in the KB. The framework is divided into six parts, including overall architecture, EMC management documents, entity recognition, entity linking, KB, and the QA system. Detailed descriptions follow.

A. Overall Architecture

The overall architecture has the following three phases:

1) blue phase: KB construction for EMC management. This phase includes the steps 1 to 3;
2) yellow phase: feature extraction of the product EMC into entities. This phase consists of steps 4 and 5; and
3) green phase: entity linking and management plan generation. This phase includes the steps 6 to 8.

Each phase includes several steps as shown in Fig. 1.

All the steps in Fig. 1 are explained as follows.

1) Basic materials to build the KB for the EMC management are obtained from the extensive and well-documented EMC design and EMI test regulations [19], [20].
2) EMC entity recognition is achieved, wherein entities are identified, and the properties are correlated with the CNN method.
3) Entities are linked and stored in the KB.
4) The design specification and instruction of target products are imported.
5) EMC entity recognition is fulfilled for the target products through the CNN method.
6) Entities and product properties are correlated into a group of sub knowledge graph for the product.
7) The QA engine ranks and translates the knowledge graphs into sorted sentences.
8) Finally, the management plan is generated and delivered to the EMC department.

B. EMC Management Documentations

The International Electrotechnical Commission (IEC) 61000 specification series and CISPR 22 for EMC are sourced as the authoritative documentations in our model. IEC 61000 series of standards are general and universal international standards for EMC. These standards are the basic standards issued by the IEC.

International Special Committee on Radio Interference (CISPR) is the IEC Commission on EMC and is responsible for the EMI radio signal protection test standard specification of all types of electrical equipment. CISPR 22 is applicable to identify radio disturbance characteristics of the information technology equipment.

C. Entity Recognitions

Named entity recognition (NER) has been an important subject for the past 20 years [21], [22]. From the early methods based on dictionaries and rules to the traditional machine learning methods, the main probabilistic models include the hidden Markov model, maximum entropy model, and conditional random field model. In recent years, NER has gradually evolved to incorporate the deep learning methods, such as the CNN and recurrent neural network. We extended the research of electronic medical records on EMC named entities [23].

Named entities fall under the following predefined categories:

1) EMC requirement specification and rank,
2) equipment assessment,
3) test requirements and validation,
4) measurements,
5) EMC design principles, and
6) EMC environment evaluation.

The CNN is trained to classify the tagged entities in IEC 61000 into the six categories. The process is shown in Fig. 2.

D. Entity Linking

After the entities are recognized and classified, entity linking is broken down to intracategory linking and intercategory linking. Entity linking can convert text data into the labeled entities, thereby allowing computers to process entities rather than the text. The TransR methodology which constructs entities and relational embedding in entity space and relational space is

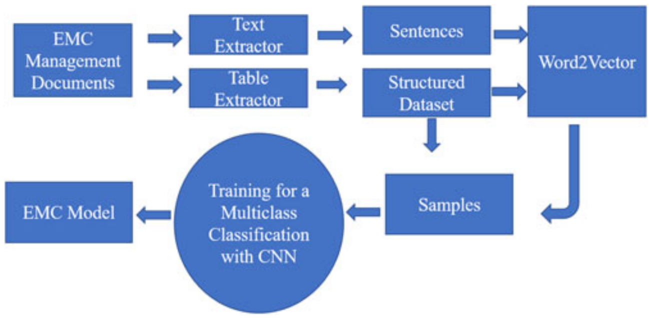

Fig. 2. EMC management process with CNN.

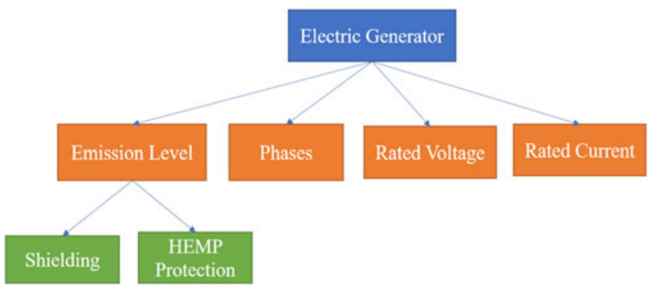

Fig. 3. Entity tree.

applied in this context [24]. The relationship between the head entity (h), relation (r), and tail entity (t) is

$$\boldsymbol{h} + \boldsymbol{r} = \boldsymbol{t}. \tag{1}$$

The score function f_r is expressed as [25]

$$\boldsymbol{f_r}\,(\boldsymbol{h}, \boldsymbol{t}) = |\boldsymbol{h} + \boldsymbol{r} - \boldsymbol{t}| \tag{2}$$

f_r is small when t is the entity that is being sought. Then, embedding is conducted. The entities are first projected into the corresponding relational space. Then, we establish the translation relationship from the first to the last entity. Finally, the entities are clustered as an entity tree. An entity tree is depicted in Fig. 3 with an electric generator as an example.

In this example, the electric generator is taken as an entity, which has four attributes as emission level, phase, rated voltage, and rated current. The attribute of emission level has two predicates as shielding and high altitude electromagnetic pulse (HEMP) protection.

E. Knowledge Base

The linked entities are stored in the repository as follows:

```
{
    "name": "electric generator",
    "category": "equipment",
    "aspects":
    {
      "aspect": ["emission level", "phases", "rated
     voltage", "rated current"]
    }
}
```

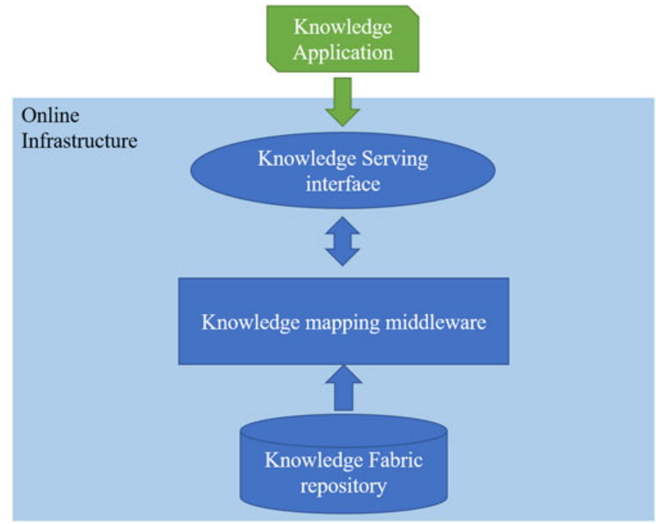

Fig. 4. Online serving model of KB.

A single row or blob of data can be projected into many nodes independent of other data.

The "create, read, update, and delete" operations are supported by the repository in an atomic manner.

As the entities are stored in a comprehensible manner, experienced EMC officers may perform the rectification on demand, particularly for the ranks in the EMC requirement specification.

The online serving model of the KB is shown in Fig. 4.

The mapping middleware is responsible for translating the entity query by normalized name and category and then mapping the result row set in the knowledge graph into a tree.

F. Product Specification and QA

Many studies have been conducted on generic QA for user generated questions [26]. However, automatically answering questions asked in natural language in a specific domain remains problematic, particularly in the EMC field, because such a process requires professional skills and experience. Therefore, applying a simplified model is necessary to achieve a question-answering framework in the EMC domain [27].

During the learning process, our model is trained to recognize the optimal KB. The specific methods are as follows:

1) Extracting unstructured information into related entities. Product specification can be identified as a set of entities by applying the model trained in Section C. Entities which contain the names, categories of entities, and certain indices adequately reflect the features and qualities of the product. Our model primarily adopts the knowledge-based question-answering (KBQA) approach and computes answers to the natural language questions or the product specification using the extant KB. Fig. 5 shows the main flowchart of the KBQA approach.
2) Transforming entities into a group of knowledge graphs. Our KB has numerous entities and relations. Thus, many subknowledge graphs are possible in the following format:

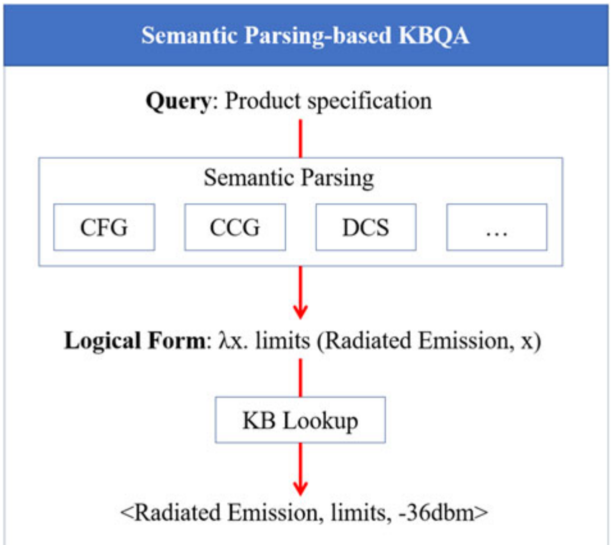

Fig. 5. Main flowchart of KBQA approaches.


```
{
    “name”: “emission level”,
    “category”: “aspect”,
    “value”:
    {
        “low”: “200 kv/m”,
        “high’: “300 kv/m”
    }
    “knowledge”:
    {
        “name”: “emission level”,
        “category”: “aspect”,
        “environments”:
        {
            “requirements”:
            {
                “requirement1”: “”,
                “requirement2”: “”
            }
        }
        “measures”:
        {
            “”:””
                “tests”:
                {
                    “”:””,
                }
    }
    “principles”
    {
        “principle1”:””,
        “principle2”:””
    }
    }
}
```


Fig. 6. Management process based on knowledge graphs.

3) By evaluating the bias of the values of the entities with the knowledge graph and requirement ranks, the knowledge graphs are sorted into a list, which is used as the material to generate sentences. Although the entities of one product may generate several graphs in Step 2, only one or a few of them will be a representation of the product. The QA engine will rank subknowledge graphs and select subgraphs with higher scores. Then, the QA system translates those subknowledge graphs into sentences [28], [29].
4) Finally, the sentences are mapped into the sections of the document template library. The management plan for this product is generated automatically and provided to the EMC officer.

In Fig. 6, first, the entities are transformed into a group of knowledge graphs. Then, the entities and their attributes are ranked according to the knowledge graphs. A preliminary document plan is generated with the ranking result. Finally, a management plan is created based on the sentences translated by the QA system.

III. Experiments

The IEC/TS 61000 series specifications focusing on the system and equipment design phases include safety requirement specification, design/development, validation, manufacture, and validation. Based on the document, we built an online KB and QA application following the methodology in Section II.

Two experiments were designed to evaluate the proposed method.

A. CNN Modeling for EMC

The CNN modeling for EMC entity recognition is the key point for KB construction. Thus, the performance of the model is crucial.

TABLE I
RESULT OF EMC ENTITY RECOGNITION

	Documentation	Tagged entities	Recall	Precision
Training set	IEC 61000 series	28,981	92.09%	88.99%
Test set	IEC 60601-1-2	13,876	90.43%	87.38%

The IEC 60601-1-2 [30] indicates the EMC requirements and test specification for medical electrical equipment, which is the perfect test document to evaluate the CNN modeling. The experiment results are listed in Table I.

The $F1$ measure combines recall (r) and precision (p) with an equal weight in the following form:

$$F1 = \frac{2rp}{r+p}. \tag{3}$$

According to the definition, p is the number of correct results divided by the number of all returned results, and r is the number of correct results divided by the number of results that should have been returned [31].

Results indicate that

1) the training set self-evaluation score is 90.5%,
2) the test set F1 score is 88.9%.

Thus, approximately 90% of entities could be recognized and classified into the correct categories.

B. End to End Evaluation

1) Test Product Specifications: The General Electric (GE) specifications are available for downloading and cover almost all electric and electronic categories [32]. For this study, ten specifications were selected as the test set.

Evaluating the quality of EMC management plans generated is difficult. Fortunately, several papers provide guidelines for perusing the IEC 61000 documents to provide EMC design expertise or testing [10].

The literature [10] provides EMC management plans based on IEC 61000. In each plan, the tester must mark the key points with scores. The sum of scores for one plan is 100, making the total score 1000.

Plans generated by the machines also mark all the sentences with identified entities. Thus, the testers can leverage the identified entities to match the key points written by humans. Results are shown in Table II.

The average score is 71.6 and the standard deviation is 23.5, which is high. The root cause of the deviation is that certain terms in low-voltage equipment specifications do not match the terms in the IEC 61000 series. Thus, the entities cannot be mapped in the KB.

TABLE II
RESULTS ACCORDING TO GE SPECIFICATIONS

Specification name	Score
16466 Low Voltage Plug-in and Feeder Busway	42
40 92 49CT Variable Frequency Drives-Constant Torque and Fan and Pump Applications	76
DET1000 EntellEon Guideform Spec	59
16140 Prefabricated Power Equipment Centers	98
16484A Electronic Soft-Start Controllers-ASTAT BP	43
26 43 13 Guideform Specification for GE Surge Protective Device (SPD)	92
16402 Low Voltage Switchboard Group Mounted-Evolution	42
16355 Medium Voltage Metal Enclosed Breaker	74
16278 Three Phase Pad-Mounted Transformers	97
16610A Guideform Specification for GE TLE Series UPS, 750 kW—1 MW	93

IV. CONCLUSION

This study proposed a bootstrap framework to build KB for EMC management. The expertise of EMC officers in different industries can be stored in the repository in the form of entities and their links. Management plans are generated automatically based on the specifications of products. This KB is chiefly achieved by KBQA which generates the management plan by locating the corresponding entities of the product specification in the repository.

The proposed method is applicable for the EMC management of industrial products and provides an original and efficient method of implementing EMC management.

For further work, we can expand the EMC standards beyond the IEC/TS 61000 series specifications to guarantee the creation of a more accurate and general management plan. In addition, we can also adopt the bidirectional long short term memory network (LSTM)-CNNs for the recognition of entities, which can allow for the rapid learning of the entities and complex relationships from massive amounts of data.

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